Exercise list R8 — Corporate governance, control rights and investor activism

Covers syllabus topics 1 (corporate governance) and 7 (activism): the separation of ownership and control, the free-rider problem and the large shareholder, board control and leverage, proxy contests, active monitoring by a large investor (Tirole Ch. 9) and the allocation of control rights between insiders and outsiders (Tirole Ch. 10; Ch. 1 has no exercises, so governance is practised through Ch. 10). 15 real exercises ordered easy → really hard, copied verbatim from the course books for personal study; pages are printed, with the PDF page in brackets. Worked solutions: `lista-r8-governance-activism-solutions.pdf` (short answers: `...-solutions-short.pdf`). Links: exercise index · course programme.

References

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Train
1	Easy	Ross *Fundamentos* 9e	Cap. 1	CQ 1.14	23 (47)	no	Mana
2	Easy	Ross *Fundamentos* 9e	Cap. 1	CQ 1.15	23 (47)	no	Own
3	Easy	Grinblatt & Titman 2e	Ch. 18	18.2	653 (666)	no	Dete
4	Easy	Grinblatt & Titman 2e	Ch. 18	18.5	653 (666)	no	Priva
5	Easy	Grinblatt & Titman 2e	Ch. 18	18.4	653 (666)	no	Boar
6	Exam	Amaro de Matos	Ch. 8	8.2.6 #10	223 (241)	yes, p. 278	Harr
7	Exam	Amaro de Matos	Ch. 8	8.1.5 #5	205 (223)	yes, pp. 271–272	Shlei
8	Exam	Tirole	Ch. 10	10.5	420 (433)	yes, p. 613	Inves
9	Exam	Tirole	Ch. 10	10.7	421 (434)	yes, pp. 613–614	Limi
10	Exam	Tirole	Ch. 10	10.6 (with 10.5 as Part 0)	420–421 (433)	yes, p. 613	Cont
11	Exam	Tirole	Ch. 9	9.1	379 (392)	yes, pp. 606–607	High
12	Really hard	Tirole	Ch. 10	10.3	419–420 (432)	yes, p. 612	Larg
13	Really hard	Tirole	Ch. 10	10.8	421–422 (434)	yes, p. 614	Cont
14	Really hard	Tirole	Ch. 10	10.9	422 (435)	yes, pp. 614–615	Cont
15	Really hard	Tirole	Ch. 10	10.2	418–419 (431)	yes, pp. 611–612	Cont

Tier 1 — Easy

1. Management opposes a hostile offer

Source: Ross, Cap. 1, Questão 14 (Revisão de conceitos e questões instigantes), p. 23 (PDF 47) · Answer in book: no

14. Problemas de agência [OA4] Suponhamos que você tenha ações de uma empresa. O preço atual por ação é $25. Outra empresa acaba de anunciar que deseja comprar a sua empresa e que pagará $35 por ação para adquirir todas as ações em circulação. A administração de sua empresa começa imediatamente a se opor a essa oferta hostil. A administração está agindo em busca do melhor interesse dos acionistas? Por quê?

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2. Ownership structure, agency problems and institutional investors

Source: Ross, Cap. 1, Questão 15 (Revisão de conceitos e questões instigantes), p. 23 (PDF 47) · Answer in book: no

15. Problemas de agência e propriedade corporativa [OA4] A propriedade corporativa varia em todo o mundo. Historicamente, indivíduos têm tido a propriedade da maioria das ações das corporações estadunidenses. Na Alemanha e no Japão, porém, os bancos, outras grandes instituições financeiras e outras empresas possuem a maioria das ações das empresas abertas. Você acha que os problemas de agência podem ser mais ou menos sérios na Alemanha e no Japão do que nos Estados Unidos? Por quê? Nos últimos anos, grandes instituições financeiras, como os fundos mútuos e os fundos de pensão, têm se tornado os proprietários dominantes de ações, e essas instituições estão mais ativas nas questões corporativas. Quais são as implicações dessa tendência para os problemas de agência e o controle corporativo?

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3. Determinants of ownership concentration

Source: Grinblatt, Ch. 18, Exercise 18.2, p. 653 (PDF 666) · Answer in book: no

18.2. Discuss the factors that determine whether firms are likely to have large ownership concentrations.

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4. A law against blockholders' private benefits

Source: Grinblatt, Ch. 18, Exercise 18.5, p. 653 (PDF 666) · Answer in book: no

18.5. As a Washington policy analyst, you are asked to comment on a proposed law that would make it more difficult for large outside shareholders to extract private benefits from the partial control they can exert over management. How would such a law affect the incentives of outside shareholders to monitor management?

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5. Board control and debt ratios

Source: Grinblatt, Ch. 18, Exercise 18.4, p. 653 (PDF 666) · Answer in book: no

18.4. Consider three similar firms that differ only in the extent to which they are controlled by their boards of directors. In firm 1, the board has complete control of the investment decisions, operating decisions, and financing choices. In firm 2, the board is unable to monitor investment and operating decisions, but they do control financing decisions. In firm 3, the board has very little control over either investment, operating, or financing decisions. Describe how debt ratios are likely to differ in the three firms.

Tier 2 — Exam level

6. Proxy contests: contenders vote for themselves

Source: Amaro de Matos, Ch. 8, Exercise 8.2.6 #10, p. 223 (PDF 241) · Answer in book: yes, p. 278

10. Show that the incumbent and the rival managements will always want to vote for themselves regardless of their private signals.

Context needed to solve it *From Amaro de Matos §8.2.4 "The Choice of Takeover Methods", pp. 214–216 (PDF 232–234):*

A paper by Harris and Raviv (1988) goes into greater depth in exploring the determinants of corporate takeover methods and their outcomes and price effects.

In their model there are three types of agent: the incumbent management, a rival management, and a large number of atomistic shareholders. To finance the \$1 investment required, the incumbent may issue common stock and (nonvoting) debt. It is assumed that the only effect of debt is to increase the fraction of votes controlled by one of the contenders. Since debt reduces the value of the incumbent's control, in the absence of a takeover threat there will be no debt.

It is assumed that the management currently in control has one of two possible levels of ability (say, 1 and 2), with probabilities k and $(1-k)$, respectively. The present value of cash flows Y depends only on the ability of the management; its two possible values are denoted, respectively, Y_1 and Y_2, with $Y_1 > Y_2$.

The expected value of the firm is

$$Y_I = kY_1 + (1-k)Y_2.$$

The incumbent is supposed to have all its wealth W_I invested in the firm. Thus the fraction of the firm owned by the incumbent is

$$\alpha = \frac{W_I^*}{Y_I} \quad \text{with } W_I^* = W_I + Y_I - 1.$$

What happens when there is a takeover threat? This model assumes that there are benefits to being in control of the firm in addition to those that originate in net cash flows. In the event that there is a rival management trying to take over, the incumbent, to the extent that he/she owns stock in the firm, should not be opposed to the takeover if he/she believes that the rival has a higher ability. Benefits from control, however, provide contenders with incentives to vote for themselves regardless of their beliefs.

When the rival appears, the incumbent may choose, as an antitakeover device, to issue debt in exchange for equity, with the entire exchange being completed before the rival purchases equity. The value of the equity for a debt level D and controlling management of ability $i = 1, 2$ is denoted by $E_i(D)$, and it is assumed that $E_1 > E_2$ and E is decreasing with D. Then the incumbent's share of the equity becomes

$$\alpha_I(D) = \alpha Y(D)/E(D)$$

where $Y(D)$ denotes the value of the firm when the face value of the debt is D. Under Harris and Raviv's assumptions given the ability of the incumbent management, the value of the firm is not affected by the debt level. In other words, the dependence of Y on D appears only because the debt level affects the determination of who controls the firm. It follows that $\alpha_I(0) = \alpha$.

Also, assuming that the rival management invests its wealth W_R in the firm's stock, and given that the incumbent already has α_I, his/her share will be

$$\alpha_R(D) = \min\{W_R/E(D),\ 1 - \alpha_I(D)\},$$

and the passive investors will retain the rest:

$$\alpha_P(D) = 1 - \alpha_R(D) - \alpha_I(D).$$

The goal of the incumbent is to maximize the value of his/her shares plus the expected benefits of control. Denoting by $K(D)$ the benefits of control to the incumbent given a debt level D, the incumbent's problem is to choose D in such a way as to maximize

$$\alpha Y(D) + K(D). \qquad (8.16)$$

Given the resulting distribution of ownership $(\alpha_P, \alpha_R, \alpha_I)$, the incumbent's share of the votes in any control contest consists of his/her own votes plus a fraction q_1 or q_2 of the passive votes, depending on his/her ability i, where $q_1 > q_2$. Here, q_i is the exogenous fraction of the passive shareholders voting for the incumbent, given his/her ability is i.

Hence, if the incumbent is of ability i, he/she will have $\alpha_I + q_i\alpha_P$ votes. It is clear that this number depends on the debt level. But on the other hand, the outcome of the takeover attempt depends on the number of votes on the side of the incumbent and therefore on the debt level choice.

For instance, a *successful tender offer* occurs in a situation in which the incumbent (even if of higher ability) will not get 50% of the votes. Formally, a tender offer is successful if D is chosen so as to satisfy

$$\alpha_I(D) + q_2\alpha_P(D) \geq \frac{1}{2}. \qquad (8.17)$$

[...] Finally, a proxy fight is a situation in which the incumbent gets more than 50% of the votes only if it is of type 1.

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7. The large shareholder: premium, cutoff and firm value

Source: Amaro de Matos, Ch. 8, Exercise 8.1.5 #5, p. 205 (PDF 223) · Answer in book: yes, pp. 271–272

5. When there is a large shareholder, explain and show what happens to the following variables as the large shareholder becomes larger:

(a) The tender offer, $\pi(\alpha)$.

(b) The cutoff value of Z, $Z^c(\alpha)$.

(c) The market value of the firm, $V(\alpha, q)$.

Assume now that $c = 0$ and let $F(Z)$ be the uniform distribution on $[0, 1]$. Solve for the values of $\pi(\alpha)$ and the cutoff value $Z^c(\alpha)$. Verify the relationships derived earlier in the exercise.

Context needed to solve it *From Amaro de Matos §8.1.2 "The Role of a Large Shareholder", pp. 196–198 (PDF 214–216):*

In the former model the bidder is an outsider who makes a profit at the expense of the shareholders, who agree with the takeover in order to punish the existing management. It is known, however, that large blockholders serve as an internal control on the managerial behavior and play an important role in takeovers. This provides an alternative defensive mechanism against bad management, through the possibility of proxy fights or tender offers from the large shareholders.

An extension of the former model seems to be in order, substituting for Assumption 8.1.1 with

Assumption 8.1.2. There is one large minority shareholder who owns a given proportion of the shares, and there is a fringe of atomistic shareholders.

This idea was formalized by Shleifer and Vishny (1986). Hereinafter the large shareholder will be referred to as L. Let α be the proportion of the shares owned by L. Since he/she is a minority, $0 < \alpha < \frac{1}{2}$. Also, let Z denote the random increase in the discounted market value generated by the substitution of an inefficient management.

Assume that the increase in value comes from the fact that L has privileged access to a new technology. Let I denote the probability of drawing a positive improvement value Z from a probability distribution $F(Z)$ with support on $(0, Z_{\max}]$. In this case, I can be thought of as research intensity. This probability (or research intensity) comes at a cost $c(I)$ satisfying $c' > 0$ and $c'' > 0$. Under these conditions, the bidder will move to make an offer if he/she can purchase $\frac{1}{2} - \alpha$ of the shares. The expected market value under the existing management being q, the tender price must be higher than q, say,

$$p = q + \pi,$$

where π must satisfy the condition

$$\frac{1}{2}Z - (0.5 - \alpha)\pi - c \geq 0, \qquad (8.2)$$

such that the bid is profitable for the raider. For the small shareholders the expected benefit from a bid is

$$E[Z \mid 0.5Z - (0.5 - \alpha)\pi - c \geq 0]$$

and therefore, to tender is the best strategy if

$$\pi - E[Z \mid Z - (1 - 2\alpha)\pi - 2c \geq 0] \geq 0. \qquad (8.3)$$

The role of L is to define the tender price by choosing the value of π that minimizes the left-hand side of (8.3). The optimal value clearly depends on α. Let it be denoted by $\pi^*(\alpha)$. Notice that $\pi^*(\alpha)$ is just the expected value of an improvement, conditional on the takeover being profitable.

Shleifer and Vishny show that $\pi^*(\alpha)$ is unique and is a decreasing function of α. Notice that $\alpha = 0$ replicates the case of diffuse ownership discussed already. In this case, equation (8.2) implies that $Z > \pi$ is a necessary condition for a takeover to succeed. But under that condition no one tenders, reproducing the result of Grossman and Hart (1990).

Defining $Z^c(\alpha)$ as the cutoff value of Z that makes L indifferent between taking over and not (that is to say, the smallest value of Z satisfying (8.2) with $\pi \equiv \pi^*(\alpha)$), they show that $Z^c(\alpha)$ is a strictly decreasing function. Also, $I^*(\alpha)$, the optimal probability of drawing a positive value of Z, increases with α, the proportion of shares held by L. [...]

To see how an increase in α affects the market value of the firm, notice that this market value may be written as

$$V(\alpha, q) = q + I^*(\alpha)\{1 - F[Z^c(\alpha)]\}E[Z \mid Z \geq Z^c(\alpha)].$$

This formula tells us that the value of the firm increases from the original value q by an amount that is proportional to the product of three factors: (1) the optimal probability of drawing a positive value of Z, (2) the expected value of increase in discounted market value generated by the substitution of an inefficient management, provided that it is above a certain minimum threshold that makes the investment worthwhile, $E[Z \mid Z \geq Z^c(\alpha)]$, and (3) the probability that such an increase in profits is above that threshold, $1 - F[Z^c(\alpha)]$. [...]

Proposition 8.1.1 (Shleifer and Vishny, 1986). An increase in the proportion of shares held by L results in a decrease in the takeover premium but an increase in the market value of the firm.

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8. When investor control makes financing harder

Source: Tirole, Ch. 10, Exercise 10.5, p. 420 (PDF 433) · Answer in book: yes, p. 613

Exercise 10.5 (when investor control makes financing more difficult to secure). The general thrust of control rights theory is that investors are reassured, and so are more willing to lend, if they have control rights over the firm. The purpose of this exercise is to build a counterexample in which investor control is self-defeating and jeopardizes financing.

(i) An entrepreneur has cash A and wants to invest $I > A$ into a (fixed-size) project. The project yields $R > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_H if the entrepreneur behaves and $p_L = p_H - \Delta p$ ($\Delta p > 0$) if the entrepreneur misbehaves. The entrepreneur receives private benefit $B > 0$ in the latter case, and 0 in the former case. All parties are risk neutral, the entrepreneur is protected by limited liability, and the rate of interest in the economy is 0.

What is the necessary and sufficient condition for the entrepreneur to be able to obtain financing from investors?

(ii) Now add a control right. This control right can raise the expected revenue in the case of misbehavior, but does nothing in the case of good behavior; namely, the holder of the control right can select an action ("damage control") that raises the probability of success from p_L to $p_L + \nu$ ($\nu > 0$) in the case of misbehavior, but keeps p_H constant. This interim action imposes a cost $\gamma > 0$ on the entrepreneur. (If the action is not selected, the probabilities of success are as in question (i), and there is no private cost γ.) The choice of action is simultaneous (say) with the entrepreneur's choice of effort.

First assume "entrepreneur control" (the entrepreneur is given the right to select this action or not). Write the two incentive constraints for the entrepreneur to behave. Show that, compared with question (i), the pledgeable income remains the same if $\nu B/(\Delta p) \leqslant \gamma$, and is decreased otherwise.

(iii) Next consider "investor control." Assume that when indifferent, the investors select the dominant strategy, i.e., the damage-control action (alternatively, one can assume that the action raises p_H as well, to $p_H + \varepsilon$, where ε is arbitrarily small). Show that the financing condition is now

$$p_H \left[R - \frac{B}{\Delta p - \nu} \right] \geqslant I - A.$$

Conclude that investor control, besides reducing NPV, may also make it more difficult for the entrepreneur to secure financing.

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9. Limited versus extended investor control

Source: Tirole, Ch. 10, Exercise 10.7, p. 421 (PDF 434) · Answer in book: yes, pp. 613–614

Exercise 10.7 (extent of control). A simple variation on the basic model of Section 10.2.1 involves a choice between limited investor control and extended investor control, rather than between entrepreneur control and investor control. Suppose, in the model of Section 10.2.1, that entrepreneur control is out of the picture (after you finish the exercise, you may want to think about a sufficient condition for this to be case), but that there are two degrees of investor control:

Limited. The action taken then increases the probability of success by $\tau_A > 0$ and inflicts cost $\gamma_A > 0$ on insiders.

Extended (investors have control over a wide set of actions). The selected action then increases the probability of success by $\tau_B > \tau_A$ and inflicts cost $\gamma_B > \gamma_A$ on insiders.

Assume that

$$\tau_A R - \gamma_A > \tau_B R - \gamma_B.$$

Find conditions under which limited or extended investor control prevails.

Context needed to solve it *From Tirole §10.2.1 "The Aghion–Bolton Model", pp. 389–391 (PDF 402–404):*

To illustrate this in the simplest possible way, let us return to the basic (fixed-size) model of Chapter 3: to finance her project, the entrepreneur must borrow the difference between the investment cost I and her net worth A. The project succeeds (and then yields R) with probability p, where $p = p_H$ if she behaves and p_L if she misbehaves (and then takes private benefit B); otherwise the project fails and yields nothing. Let us further introduce the possibility of taking an interim action that

(i) raises the probability of success uniformly by $\tau > 0$ (so the probability of success becomes $p_H + \tau$ or $p_L + \tau$, depending on the entrepreneur's behavior, if the action is taken, and remains p_H or p_L if the status quo action is selected); and

(ii) engenders private cost $\gamma > 0$ for the entrepreneur (or, more generally, the firm's insiders).

[…] We assume that this interim action cannot be contracted upon at the initial (financing) stage. By contrast, the parties can contract on who is entitled to decide. […]

Figure 10.1 (timeline, left to right): *Financing stage* — project costs I; entrepreneur has equity $A < I$; borrows $I - A$. → *Interim action* — choice between status quo action (probability of success is p), and profit-enhancing action (probability of success is $p + \tau$). → *Moral-hazard stage* — entrepreneur's choice affects the probability of success: $p = p_H$ (no private benefit) or p_L (private benefit B). → *Outcome stage* — verifiable profit: R with probability p (or $p + \tau$), 0 with probability $1 - p$ (or $1 - p - \tau$).

The assumption that the profit-enhancing action is orthogonal to managerial moral hazard, i.e., raises the probability of success uniformly, simplifies the analysis since it does not affect the incentive compatibility condition: if the profit-enhancing action is to be taken, then the incentive constraint becomes

$$[(p_H + \tau) - (p_L + \tau)]R_b \geqslant B \quad \text{or} \quad (p_H - p_L)R_b \geqslant B,$$

where R_b is the borrower's reward in the case of success. Note that τ does not enter the incentive constraint. […]

Let us first focus on the interesting case in which the profit-enhancing action reduces aggregate welfare and is thus first-best suboptimal:

$$\tau R < \gamma.$$

Suppose that the control right is given to the investors. Because they share part of the profit and bear none of the cost, they indeed select the profit-enhancing action, resulting in pledgeable income:

$$(p_H+\tau)\left(R-\frac{B}{(p_H+\tau)-(p_L+\tau)}\right)=(p_H+\tau)\left(R-\frac{B}{\Delta p}\right).$$

Because investors do not earn supranormal profits in a competitive capital market (they break even), the NPV is also the entrepreneur's welfare when raising funds. This NPV must account for both the increase τ in the probability of success and the entrepreneur's cost γ associated with the interim action:

$$U_b=\text{NPV}=(p_H+\tau)R-I-\gamma.$$

[…] There is, however, no renegotiation since the entrepreneur has no money to compensate investors for the loss of value on their claims.

Suppose in contrast that the entrepreneur does not relinquish control. Because $R_b \leqslant R$, $\tau R_b < \gamma$, and therefore the entrepreneur does not pick the profit-enhancing action. In words, the entrepreneur bears the entire cost and gets only part of the benefits of the profit-enhancing action. The pledgeable income is, as in Chapter 3,

$$p_H\left(R-\frac{B}{\Delta p}\right),$$

and the NPV (i.e., the entrepreneur's payoff) is

$$p_H R-I>(p_H+\tau)R-I-\gamma.$$

As expected, allocating control to investors reduces the NPV by $\gamma-\tau R>0$. But it increases pledgeable income by $\tau[R-(B/\Delta p)]$.

Suppose now that

$$p_H\left(R-\frac{B}{\Delta p}\right)<I-A<(p_H+\tau)\left(R-\frac{B}{\Delta p}\right).$$

Then the entrepreneur has insufficient cash on hand and can raise funds only by relinquishing the control right to the investors. […]

Lastly, we note that the allocation of control is a trivial issue when investor control is first-best optimal, that is, when

$$\tau R>\gamma.$$

In this case, investor control increases both the NPV from $p_H R-I$ to $(p_H+\tau)R-I-\gamma$ and the pledgeable income from $p_H(R-B/\Delta p)$ to $(p_H+\tau)(R-B/\Delta p)$.

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10. Control and incentives: complements or substitutes

Source: Tirole, Ch. 10, Exercise 10.6, pp. 420–421 (PDF 433) · Answer in book: yes, p. 613

Exercise 10.6 (complementarity or substitutability between control and incentives). This exercise pursues the agenda set in Exercise 10.5 by considering various forms of complementarity and substitutability between the exercise of control rights and managerial incentives. It therefore relaxes the assumption of separability between the two.

(i) An entrepreneur has cash A and wants to invest $I>A$ into a (fixed-size) project. The project yields $R>0$ with probability p and 0 with probability $1-p$. The probability of success is p_H if the entrepreneur behaves and $p_L=p_H-\Delta p$ ($\Delta p>0$) if the entrepreneur misbehaves. The entrepreneur receives private benefit $B>0$ in the latter case, and 0 in the former case. All parties are risk neutral, the entrepreneur is protected by limited liability, and the rate of interest in the economy is 0.

What is the necessary and sufficient condition for the entrepreneur to be able to obtain financing from investors?

(ii) Now consider the possibility that a profit-enhancing action be chosen. For reasons of simplicity (but not for the sake of realism!), assume that this action is chosen simultaneously with effort. This action raises the probability of success to

- $p_H + \tau_H$ if the entrepreneur behaves, and
- $p_L + \tau_L$ if the entrepreneur misbehaves.

The action is indeed profit enhancing ($\tau_L, \tau_H > 0$) and is

- complementary with effort if $\Delta\tau \equiv \tau_H - \tau_L > 0$,
- substitutable with effort if $\Delta\tau < 0$.

The action further inflicts a disutility γ on the manager, where

$$\max(\tau_L, \tau_H) \cdot R < \gamma.$$

Lastly, assume that the high effort must be induced in order for financing to occur.

Write the pledgeable income under investor control and entrepreneur control. When does investor control increase the pledgeable income (and therefore facilitate financing)?

Context needed to solve it Part 0 — the exercise whose agenda this one pursues. *From Tirole, Ch. 10, Exercise 10.5, p. 420 (PDF 433) · Answer in book: yes, p. 613 (it is also exercise 8 of this list):*

Exercise 10.5 (when investor control makes financing more difficult to secure). The general thrust of control rights theory is that investors are reassured, and so are more willing to lend, if they have control rights over the firm. The purpose of this exercise is to build a counterexample in which investor control is self-defeating and jeopardizes financing.

(i) An entrepreneur has cash A and wants to invest $I > A$ into a (fixed-size) project. The project yields $R > 0$ with probability p and 0 with probability $1 - p$. The probability of success is p_H if the entrepreneur behaves and $p_L = p_H - \Delta p$ ($\Delta p > 0$) if the entrepreneur misbehaves. The entrepreneur receives private benefit $B > 0$ in the latter case, and 0 in the former case. All parties are risk neutral, the entrepreneur is protected by limited liability, and the rate of interest in the economy is 0.

What is the necessary and sufficient condition for the entrepreneur to be able to obtain financing from investors?

(ii) Now add a control right. This control right can raise the expected revenue in the case of misbehavior, but does nothing in the case of good behavior; namely, the holder of the control right can select an action ("damage control") that raises the probability of success from p_L to $p_L + \nu$ ($\nu > 0$) in the case of misbehavior, but keeps p_H constant. This interim action imposes a cost $\gamma > 0$ on the entrepreneur. (If the action is not selected, the probabilities of success are as in question (i), and there is no private cost γ.) The choice of action is simultaneous (say) with the entrepreneur's choice of effort.

First assume "entrepreneur control" (the entrepreneur is given the right to select this action or not). Write the two incentive constraints for the entrepreneur to behave. Show that, compared with question (i), the pledgeable income remains the same if $\nu B/(\Delta p) \leqslant \gamma$, and is decreased otherwise.

(iii) Next consider "investor control." Assume that when indifferent, the investors select the dominant strategy, i.e., the damage-control action (alternatively, one can assume that the action raises p_H as well, to $p_H + \varepsilon$, where ε is arbitrarily small). Show that the financing condition is now

$$p_H \left[R - \frac{B}{\Delta p - \nu} \right] \geqslant I - A.$$

Conclude that investor control, besides reducing NPV, may also make it more difficult for the entrepreneur to secure financing.

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11. Public debt, bank debt and junk bonds

Source: Tirole, Ch. 9, Exercise 9.1, p. 379 (PDF 392) · Answer in book: yes, pp. 606–607

Exercise 9.1 (low-quality public debt versus bank debt). Consider the model of Section 9.2.1, except that the project has a positive NPV even if the entrepreneur misbehaves.

As usual, the entrepreneur is risk neutral and protected by limited liability. She has assets A and must finance an investment of fixed size $I > A$. The project yields R in the case of success and 0 in the case of failure. The probability of success is p_H if the entrepreneur behaves (no private benefit) and p_L if she misbehaves (private benefit B). Investors are risk neutral and demand a 0 rate of return.

Instead of assuming that the project has positive NPV only in the case of good behavior, suppose that

$$p_H R > p_L R + B > I.$$

Suppose further that there is a competitive supply of monitors and abundant monitoring capital. At private cost c, a monitor can reduce the entrepreneur's private benefit of misbehavior from B to b. Assume that

$$p_H \frac{B-b}{\Delta p} > c > (\Delta p)R - p_H \frac{b}{\Delta p}$$

and

$$(\Delta p)R > c + B.$$

Show that there exist thresholds $A_1 < A_2 < A_3$ such that

- if $A \geqslant A_3$, the firm issues high-quality public debt (public debt that has a high probability of being repaid);
- if $A_3 > A \geqslant A_2$, the firm borrows from a monitor (and from uninformed investors);
- if $A_2 > A \geqslant A_1$, the firm issues junk bonds (public debt that has a low probability of being repaid);
- if $A_1 > A$, the firm does not invest.

Context needed to solve it *From Tirole §9.2.1 "Benefit of Activism", pp. 356–358 (PDF 369–371):*

To model the collection of prospective information, we start from the fixed-investment model of Section 3.2 and add a monitor who can intervene in order to reduce the scope for moral hazard. A risk-neutral entrepreneur with wealth A has a project costing $I > A$ and must therefore borrow $I - A$ from investors. The project yields R when it succeeds and 0 when it fails. The probability of success is p_H if the entrepreneur works and $p_L = p_H - \Delta p$ if she shirks.

Table 9.1

	good project	bad project	Bad project
Pr(success)	p_H	p_L	p_L
Private benefit	0	b	B

9.2.1.1 No Monitoring. In the absence of monitoring, shirking provides private benefit B. Letting R_b denote the entrepreneur's reward in the case of success (she receives nothing in the case of failure as she is protected by limited liability), incentive compatibility requires that

$$(\Delta p)R_b \geqslant B. \qquad (9.1)$$

Funding requires that the pledgeable income exceed the investors' investment:

$$p_H \left(R - \frac{B}{\Delta p}\right) \geqslant I - A. \qquad (9.2)$$

If this condition is satisfied, and given that the investors break even, the entrepreneur's utility is equal to the project's NPV:

$$U_b = p_H R - I. \qquad (9.3)$$

9.2.1.2 Monitoring (With Fixed Intensity). Let us now formalize the idea that monitoring reduces the extent of moral hazard. A straightforward way of doing so is to assume that a monitor can reduce the private benefit that can be enjoyed by the entrepreneur by shirking from B to $b < B$. The monitor must, however, bear an unobservable private monitoring cost $c > 0$ in order to achieve this reduction in private benefit.

An interpretation of this monitoring structure is as described in Table 9.1. The manager will have to choose among a number of ex ante identical projects. The manager privately learns the payoffs attached to each project. There are three relevant projects: (1) the "good project," which yields no private benefit and has probability of success p_H; (2) the low-private-benefit "bad project," which yields private benefit b and has probability of success p_L; and (3) the high-private-benefit "Bad project," which yields private benefit B and has probability of success p_L.

The monitor moves first. If he incurs effort cost c, he is able to identify the high-private-benefit Bad project and thus to prevent the entrepreneur from selecting it. But he still cannot tell the other two projects apart, and so the entrepreneur, who can condition her choice of project on the existence or absence of monitoring, can still choose the low-private-benefit bad project if she wishes to. The monitor learns nothing when he does not incur the monitoring cost c; then, because the projects are still indistinguishable by the investors, the entrepreneur can choose any of the three projects, as in the absence of monitoring [...].

Let us assume that the entrepreneur "hires" a monitor and that the monitor's incentives induce him to monitor. The entrepreneur's private benefit from shirking is then equal to b, and so, if R_b denotes the entrepreneur's reward in the case of success, the entrepreneur works if and only if

$$(\Delta p) R_b \geqslant b. \qquad (9.4)$$

We can further assume that $(\Delta p) R_b < B$; for, if $R_b \geqslant B/\Delta p$, the entrepreneur is induced to work even in the absence of monitoring. Monitoring is then useless.

The monitor too must be provided with an incentive scheme. We maintain the assumption of universal risk neutrality, and so there is no loss of generality in assuming that the monitor receives a reward R_m in the case of success and 0 in the case of failure (because of limited liability). When not incurring cost c, the monitor is unable to prevent the entrepreneur from shirking, and so the incentive for monitoring is provided by an R_m satisfying

$$(\Delta p) R_m \geqslant c. \qquad (9.5)$$

Abundance of monitoring capital assumption. Let us first assume that monitoring capital is "abundant" or "not scarce." This means that there is a large supply of monitors who are willing to invest their capital in the monitoring activity as long as they are as well-off by doing so as with any other investment. They are thus willing to contribute to the firm's investment at level I_m such that

$$p_H R_m - c = I_m. \qquad (9.6)$$

The monitor then obtains no rent, and receives net payment $(p_H R_m - I_m)$ equal to his monitoring cost c. [...]

Nonmonitoring or uninformed investors are willing to fund the project if and only if

$$p_H (R - R_b - R_m) \geqslant I - A - I_m. \qquad (9.7)$$

And so, using (9.4)–(9.6), the necessary and sufficient condition for the project to be funded is

$$p_H \left(R - \frac{b}{\Delta p} \right) \geqslant I - A + c. \qquad (9.8)$$

That is, monitoring reduces the agency cost from $p_H B/\Delta p$ to $p_H b/\Delta p$, but adds monitoring cost c. Using (9.5) and (9.6), the monitor's stake R_m can be chosen equal to $c/(\Delta p)$ and the monitor's investment contribution equal to

$$I_m = \frac{p_L c}{\Delta p}.$$

[…] When does the entrepreneur benefit from having a monitor? Because all investors, including the monitor, obtain no rent from their relationship with the firm, the entrepreneur's utility is equal to the project's NPV under monitoring:

$$U_b = p_H R - I - c. \qquad (9.10)$$

Tier 3 — Really hard

12. A large minority blockholder

Source: Tirole, Ch. 10, Exercise 10.3, pp. 419–420 (PDF 432) · Answer in book: yes, p. 612

Exercise 10.3 (large minority blockholding). Consider the active monitor model (see Chapter 9). The firm yields R in the case of success and 0 in the case of failure. The entrepreneur, large shareholder, and small shareholders have shares s_1, s_2, and s_3, respectively, where $s_1 + s_2 + s_3 = 1$. (To complete the model's description, one can, as in Chapter 9, assume that $s_1 R \geqslant b/\Delta p$ and $s_2 R \geqslant c/\Delta p$, using the notation of this chapter.) The small shareholders have formal control (one share bears one voting right and $s_3 > \frac{1}{2}$).

The project can be modified in a countable number of ways ($k = 0, 1, ...$). Option 0 consists in "not modifying the project" (this option is known to everyone). Options 1 through ∞ do modify the project; all but two of them have disastrous consequences for all parties (so taking a modification at random is dominated by the status quo option 0). The two relevant modifications are such that one increases the probability of success by $\tau > 0$ and the other reduces it by $\mu > 0$. One involves a private cost $\gamma > 0$ or a private benefit $-\gamma$ for the entrepreneur, with $(\tau + \mu)s_1 R < \gamma$, and the other no such cost. Lastly, an action may involve a private benefit ξ for the large blockholder (or one of his subsidiary). There are three states of nature, as shown in Table 10.2. In each state of nature, the left-hand payoffs correspond to the (uninformed) investor-friendly modification and the right-hand payoffs to the (uninformed) investor-hostile modification.

Table 10.2 Probabilities: β (state 1); $(1-\beta)\kappa$ (state 2); $(1-\beta)(1-\kappa)$ (state 3).

	State 1 (friendly)	State 1 (hostile)	State 2 (friendly)	State 2 (hostile)	State 3 (fri
Impact on probability of success	τ	$-\mu$	τ	$-\mu$	τ
Private cost for entrepreneur	0	$-\gamma$	0	γ	0
Private benefit for large blockholder	0	0	0	0	0

The timing goes as follows:

(1) The entrepreneur learns the two relevant modifications and their impact on payoffs, and makes a proposal to shareholders.

(2) The large blockholder learns the relevant modifications and their impact on payoffs, and either seconds the entrepreneur's recommendation for a modification or makes a counterproposal.

(3) Majority shareholders decide between the status quo and the recommendation(s).

(i) Predict the outcome in each state of nature.

(ii) Add a fourth state of nature in which the entrepreneur and large shareholder see eye-to-eye and both prefer a value-decreasing action (say, the payoffs in state 4 are as in state 2, except that for the second relevant action, "γ" becomes "$-\gamma$" and the large shareholder receives ξ). What would you predict?

Context needed to solve it *From Tirole §9.2.1.2 "Monitoring (With Fixed Intensity)", pp. 357–358 (PDF 370–371) — the active monitor model the exercise refers to:*

A straightforward way of doing so is to assume that a monitor can reduce the private benefit that can be enjoyed by the entrepreneur by shirking from B to $b < B$. The monitor must, however, bear an unobservable private monitoring cost $c > 0$ in order to achieve this reduction in private benefit. […] There are three relevant projects: (1) the "good project," which yields no private benefit and has probability of success p_H; (2) the low-private-benefit "bad project," which yields private benefit b and has probability of success p_L; and (3) the high-private-benefit "Bad project," which yields private benefit B and has probability of success p_L.

[…] The entrepreneur's private benefit from shirking is then equal to b, and so, if R_b denotes the entrepreneur's reward in the case of success, the entrepreneur works if and only if

$$(\Delta p)R_b \geqslant b. \qquad (9.4)$$

[…] the monitor receives a reward R_m in the case of success and 0 in the case of failure (because of limited liability). When not incurring cost c, the monitor is unable to prevent the entrepreneur from shirking, and so the incentive for monitoring is provided by an R_m satisfying

$$(\Delta p)R_m \geqslant c. \qquad (9.5)$$

Note (list editor, not from the book): with shares s_1 and s_2 of a profit R, the rewards are $R_b = s_1 R$ and $R_m = s_2 R$; this is how the statement's conditions $s_1 R \geqslant b/\Delta p$ and $s_2 R \geqslant c/\Delta p$ map into (9.4) and (9.5).

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13. Uncertain managerial horizon and control rights

Source: Tirole, Ch. 10, Exercise 10.8, pp. 421–422 (PDF 434) · Answer in book: yes, p. 614

Exercise 10.8 (uncertain managerial horizon and control rights). This exercise considers the allocation of control between investors and management when the entrepreneur has an uncertain horizon.

We consider the fixed-investment model. The investment cost is I and the entrepreneur has only $A < I$. The entrepreneur is risk neutral and protected by limited liability; the investors are risk neutral and demand rate of return equal to 0. The profit is equal to R in the case of success and is 0 in the case of failure. In the absence of profit-enhancing action, the probability of success is p; when the profit-enhancing action is taken this probability becomes $p + \tau$, where $\tau > 0$, but the action imposes a nonmonetary cost on insiders, γ, where

$$\gamma > \tau R.$$

As usual, $p = p_H$ if the entrepreneur behaves (no private benefit) and $p = p_L$ if she misbehaves (private benefit B).

The twist relative to Chapter 10 is that the entrepreneur may not be able to run the project to completion: with probability λ, she must quit the firm for exogenous reasons. She learns this after the investment is sunk, but before the moral-hazard stage. If the entrepreneur quits (which will have probability λ), a new and *cashless* manager will be brought in. This manager is also risk neutral and protected by limited liability and has the same private benefit, probabilities of success, and payoff in the case of success as the entrepreneur.

Figure 10.10 summarizes the timing.

Figure 10.10 (timeline): *Date 0* — project costs I; entrepreneur has $A < I$. → *Realization of entrepreneur's horizon*: with probability $1 - \lambda$ the entrepreneur stays; with probability λ a new manager is appointed. → *Date 1* (on both branches) — moral hazard; exercise of control right (profit-enhancing action?). → Success R with probability p, or $p + \tau$, or failure (0).

Let x and y in $[0, 1]$ denote the probabilities that investors receive control when the entrepreneur and the replacement manager are in charge, respectively. And assume that

$$(p_H + \tau)\frac{B}{\Delta p} \geqslant \gamma$$

(interpret this assumption), and that

$$\rho_1 \equiv p_H R > I > \rho_0^+ \equiv (p_H + \tau)\left(R - \frac{B}{\Delta p}\right).$$

(i) Assuming that incentives must be provided for good behavior (by either the entrepreneur or the replacement manager), write down the following.

- The entrepreneur's utility. (Hint: this utility is slightly different from the project's social value. Why?)
- The pledgeable income and the breakeven condition.

(ii) Argue that $y = 1$. Find the conditions under which the project is undertaken. (Warning. Two conditions must be fulfilled: investors must be willing to finance it, and the entrepreneur must be willing to go ahead with it.)

Context needed to solve it *From Tirole §10.2.1 "The Aghion–Bolton Model", pp. 389–390 (PDF 402–403) — what investor control and entrepreneur control do in "Chapter 10":*

[…] if the profit-enhancing action is to be taken, then the incentive constraint becomes

$$[(p_H + \tau) - (p_L + \tau)]R_b \geqslant B \quad \text{or} \quad (p_H - p_L)R_b \geqslant B,$$

where R_b is the borrower's reward in the case of success. Note that τ does not enter the incentive constraint. […]

Suppose that the control right is given to the investors. Because they share part of the profit and bear none of the cost, they indeed select the profit-enhancing action, resulting in pledgeable income:

$$(p_H + \tau)\left(R - \frac{B}{(p_H + \tau) - (p_L + \tau)}\right) = (p_H + \tau)\left(R - \frac{B}{\Delta p}\right).$$

[…] Suppose in contrast that the entrepreneur does not relinquish control. Because $R_b \leqslant R$, $\tau R_b < \gamma$, and therefore the entrepreneur does not pick the profit-enhancing action. In words, the entrepreneur bears the entire cost and gets only part of the benefits of the profit-enhancing action. The pledgeable income is, as in Chapter 3,

$$p_H\left(R - \frac{B}{\Delta p}\right),$$

and the NPV (i.e., the entrepreneur's payoff) is

$$p_H R - I > (p_H + \tau)R - I - \gamma.$$

From Tirole §10.2.1, footnote 10, p. 390 (PDF 403) — random allocation of control:

Optimally, the entrepreneur would want to relinquish control stochastically under the set of inequalities just stated: that is, to give control to investors with probability x, and to retain control with probability $1 - x$, such that $(p_H + x\tau)(R - B/\Delta p) = I - A$.

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14. A continuum of control rights

Source: Tirole, Ch. 10, Exercise 10.9, p. 422 (PDF 435) · Answer in book: yes, pp. 614–615

Exercise 10.9 (continuum of control rights). This exercise extends the analysis of Section 10.2.2 to a continuum of control rights. As in Section 10.2.2, consider a risk-neutral entrepreneur protected by limited liability. The entrepreneur has cash on hand A and wants to finance a project with cost $I > A$. The project yields R if it succeeds and 0 if it fails. Investors are risk neutral and demand a rate of return equal to 0. There is a continuum of control rights, where the decision attached to a control right can be thought of as a modification relative to the initial project and is characterized by the pair (t, g): $t \gtrless 0$ is the increase in the probability of success and $g \gtrless 0$ is the private cost borne by the entrepreneur if the decision is taken (the modification is made). Let $F(t, g)$ denote the continuous joint distribution over the space of control rights and $E_F[\cdot]$ the expectations with respect to distribution F.

The probability of success is

$$p + \tau \equiv p + E_F[t\, x(t, g)],$$

where $x(t, g) = 1$ if the decision (t, g) is taken and 0 otherwise. Similarly, let

$$\gamma \equiv E_F[g\, x(t, g)].$$

Moral hazard is modeled in the usual way: $p = p_H$ if the entrepreneur behaves (no private benefit) and $p = p_L$ if the entrepreneur misbehaves (and receives private benefit B). Assume that the project can be funded only if the entrepreneur is provided with the incentive to behave.

(i) Solve for the optimal policy $x(\cdot, \cdot)$, assuming that the investors' breakeven constraint is binding (which it is for A small enough or I large enough).

(ii) Show that, as A decreases, τ and γ increase.

(iii) Discuss the implementation of the optimal $x(\cdot, \cdot)$ function.

(iv) Consider the degenerate case in which g is the same for all control rights ($g > 0$). Show that

$$\frac{d^2\gamma}{d\tau^2} > 0.$$

Context needed to solve it *From Tirole §10.2.2 "Multiple Control Rights", pp. 392–393 (PDF 405–406):*

In practice, there are multiple control rights to be divided between insiders and outsiders: product design, day-to-day management, long-term strategic decisions, hiring decisions, mergers, alliance building, etc. The analysis above is straightforwardly generalized. The intuition derived from the single-control-right analysis suggests, and this section confirms, that, in the presence of multiple control rights, it is always optimal for the entrepreneur to abandon all rights for which investor control is first-best optimal as well as, possibly, some rights for which it is not. Again, the optimal split of rights accounts not only for the value (NPV) impact of the allocation, but also for its impact on pledgeable income.

More formally, and generalizing the framework of Section 10.2.1 to multiple rights, suppose there are K dimensions of decision making and therefore K rights to allocate. Each right $k \in \{1, \ldots, K\}$ is characterized by the uniform increase $\tau_k > 0$ in the probability of success and the cost $\gamma_k > 0$ borne by insiders if the course of action is altered in dimension k. The governance structure is now defined by the allocation $x \equiv \{x_1, \ldots, x_K\}$ of formal control rights, where $x_k = 1$ if investors obtain the control of decision k and $x_k = 0$ if the entrepreneur retains control.

Let us maximize the NPV (that is, the entrepreneur's utility) subject to being able to secure financing and to the entrepreneur's incentive compatibility constraint:

$$\max_{\{R_b, x\}} \left[\left(p_H + \sum_k \tau_k x_k \right) R - I - \sum_k \gamma_k x_k \right]$$

$$\text{s.t.} \quad \left(p_H + \sum_k \tau_k x_k\right)[R - R_b] \geqslant I - A, \qquad R_b \geqslant \frac{B}{\Delta p}.$$

The solution to this program has the following features in the interesting case in which the financing constraint is binding: there exists a threshold $\theta < 1$ such that investors receive control over decision k if and only if their relative willingness to pay for that right exceeds the threshold, that is, if and only if

$$\frac{\tau_k R}{\gamma_k} \geqslant \theta.$$

Footnote 15, p. 392 (PDF 405): Let μ denote the shadow price of the financing constraint. In the interesting case in which μ is strictly positive, then $R_b = B/\Delta p$, and the derivative of the Lagrangian with respect to x_k is equal to $[\tau_k R - \gamma_k] + \mu\tau_k[R - B/\Delta p]$. The first term is just the (first-best) efficiency of taking the profitability-enhancing stance on decision k, while the second term reflects the benefits in terms of relaxing the financing constraint. [...]

Footnote 17, p. 392 (PDF 405): [...] The only rights for which the allocation is not first-best efficient (i.e., those with $\theta \leqslant \tau_k R/\gamma_k < 1$) are rights that are given to investors. Let us first assume that the entrepreneur has committed her cash on hand A and borrowed only what is needed to fund the investment cost, $I - A$ (it turns out to be optimal to do so). Then the entrepreneur no longer has any means of payment after securing financing, the entrepreneur has no way to compensate the investors for relinquishing these rights. In other words, there are potential gains from trade ex post, but these gains cannot be reaped in the absence of compensating transfer (technically, utility is "nontransferable"). [...] We conclude that the optimal allocation of rights obtained above is not renegotiated.

From p. 393 (PDF 406): The entrepreneur's incentive constraint is then binding ($R_b = B/\Delta p$). Intuitively, the entrepreneur prefers to pay investors in the "efficient" currency (cash, which is one-for-one) than in the "inefficient" one (control rights, whose transfer involves a deadweight loss). [...] An important implication of this analysis is that, ceteris paribus, firms with stronger balance sheets (say, with a higher A; see Section 3.2.2 for a broader definition of balance-sheet strength) abandon fewer rights.

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15. Allocation of control over liquidation

Source: Tirole, Ch. 10, Exercise 10.2, pp. 418–419 (PDF 431) · Answer in book: yes, pp. 611–612

Exercise 10.2 (allocation of control and liquidation policy). This exercise considers the allocation of a control right over liquidation. As described in Figure 10.9, the framework has three dates: date 0 (financing and investment), date 1 (choice of liquidation), and date 2 (payoff in the case of continuation). There is moral hazard in the case of continuation. As usual, there is universal risk neutrality, the entrepreneur is protected by limited liability, and the investors demand a rate of return equal to 0.

Figure 10.9 (timeline): *Date 0* — financing (investors contribute $I - A$). → *Date 1* — either liquidation (verifiable liquidation value L; entrepreneur obtains $U_b \geqslant 0$ in alternative job) or continuation. → After continuation: moral hazard (choice of $p \in \{p_L, p_H\}$). → *Date 2* — profit (R with probability p, 0 with probability $1 - p$).

One will assume that the variables (p_L, p_H, R, B) in the case of continuation are known *ex ante*. As usual, misbehaving (choosing probability p_L) yields a private benefit $B > 0$ to the entrepreneur. Let

$$\rho_0 \equiv p_H\left(R - \frac{B}{\Delta p}\right)$$

and

$$\rho_1 \equiv p_H R.$$

In contrast, the liquidation proceeds L and the fallback option U_b^0 for the entrepreneur may be *ex ante* random, even though they become common knowledge at date 1 before the liquidation decision. Lastly, L is fully pledgeable to investors while none of U_b^0 is.

(i) Solve for the optimal complete (state-contingent) contract, assuming that a court is able to directly verify $\omega \equiv (L, U_b^0)$ (and the profit in the case of success) and to enforce the contract specifying the probability of continuation $x(\omega) \in [0,1]$ and the allocation of L and R between the investors and the entrepreneur.

(ii) Assume from now on that,

$$\text{for all } \omega, \quad U_b^0 \leqslant \rho_1 - \rho_0.$$

That is, in the absence of a "golden parachute" given to the entrepreneur in the case of liquidation, the entrepreneur always prefers to continue. Compare the sets Ω^{FB} and Ω^{SB} of states of nature in which continuation is optimal in the absence and presence of financing constraint. How does Ω^{SB} vary with the entrepreneur's net worth A? (A diagram will help.)

(iii) From now on, assume that the court observes neither L nor U_b^0. Only the entrepreneur and the investors do. The remaining questions look at how far one can go toward the implementation of the optimal full-observability contract described in (i) using a simple allocation of the control right concerning liquidation.

One will focus on the case in which Ω^{SB} (see question (ii)) is strictly included in Ω^{FB}, and so inefficient liquidation is required.

Suppose first that the entrepreneur has the control right and that renegotiation occurs once ω is realized. Argue that

$$\Omega^{EN} = \Omega^{FB},$$

where Ω^{EN} is the set of states of nature over which continuation occurs under entrepreneur control. Conclude that the project is then not financed.

(iv) *Investor control.* Perform the analysis of question (iii) in the case of investor control in the absence of a golden parachute (the initial contract does not provide for any compensation for the entrepreneur in the case of liquidation). Suppose that the entrepreneur does not keep any savings. Show that

$$\Omega^{IN} \subset \Omega^{SB},$$

where Ω^{IN} is the set of states of nature over which continuation occurs under investor control. Is the project financed?

(v) *Investor control with golden parachute.* Argue that a positive golden parachute ($r_b > 0$ given to the entrepreneur in the case of liquidation) is optimal when investors have control.

Context needed to solve it *From Tirole §3.2.1–3.2.2 "The Fixed-Investment Model" and "The Lenders' Credit Analysis", pp. 115–117 (PDF 128–130) — the "as usual" moral-hazard stage:*

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") […].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. […] Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. […] In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. […]

[…] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$
